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On the accuracy evaluation and correction of global single-beam depths

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ABSTRACT

Historical single-beam depths provide crucial information of the seafloor topography. This study gathers 119 million data points from 5464 global single-beam track-lines provided by the NCEI. The accuracy of the single-beam depths was evaluated by analyzing crossover errors and comparison with the GEBCO_2023 model. The results indicate that the MAD of the external crossover error in the global single-beam track-lines was 44.4 m, with the highest quality in the Atlantic Ocean (34.1 m) and the lowest quality in the Indian Ocean (117.9 m). The mean difference between global single-beam depths and the GEBCO_2023 model was -54.1 m, with a STD of 260.9 m. Data collected before 1970 generally exhibited lower quality. There was at least one error in 2623 track-lines (48% of the total data). Furthermore, 50 track-lines were affected by scale factor bias and traveling time errors. We developed a visualizing window tool to identify and correct these errors. The error-corrected single-beam dataset was obtained by eliminating individual error points, modifying the scale factor, and correcting the traveling time error. The mean difference between the error-corrected dataset and the GEBCO_2023 model is 6.8 m, with a STD of 68.5 m. This indicates a notable enhancement in the overall accuracy.

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Crossover error; ocean mapping; ship soundings; single-beam depth

Introduction

Oceans cover approximately 71% of the Earth's surface, and seafloor topography data possesses significant applications in various fields of the Earth sciences (Weatherall et al. 2015). For example, seafloor topography is crucial in uncovering and progressing plate tectonic theories (Hess 1962; Smith and Sandwell 1997), helping to reveal the geologic activity of seamounts and ocean basins (Matthews et al. 2011; Müller et al. 1997), and studying ocean circulation and climate models (Kunze and Smith 2004;

Munk and Wunsch 1998). In addition, seafloor topography plays an important role in practical applications such as navigation and infrastructure engineering (Smith 1998). Due to the inefficiency of electromagnetic waves in the water, people generally carry out seafloor topography measurements based on the principle of echolocation. Echo sounder systems have been developed since the 1940s, and this technology reflects the undulation of seafloor topography by detecting the two-way propagation time of sound waves from the transducer to the bottom and then calculating the water depth in combination with the speed of sound (Mayer 2006).

Despite conducting ship-based single-beam echo measurements for decades, the measurement mechanism of single-beam echo sounders, the slow speed of ships and the economic reasons have limited the investigation of seafloor topography in considerable areas of the oceans (Mayer 2006; Tozer et al. 2019), with substantial gaps between surveying lines (as shown in Figure 1). Currently, less than 25% of the ocean floor has been mapped using shipborne sounding technology (Harper and Sandwell 2024). The objective of the Seabed 2030 program is to create a comprehensive worldwide bathymetry database with a high resolution (800–100 m) by the year 2030. To achieve complete coverage by that time, it will take hundreds of ship-years of effort using current technology (Mayer et al. 2018). The precise mapping of the worldwide seafloor topography is a challenging task that cannot be accomplished in the short term, while there is still a large amount of historical single-beam data that is worth utilizing.

The application of mapping the seafloor using single-beam echo sounders became commonplace after the Second World War (Mayer 2006), and over the course of 70 years, a large amount of single-beam data has been collected. However, the uncertainty of echo sounder measurements affects the accuracy of acoustic-based seafloor topography (Lecours et al. 2016). The lack of the GNSS (Global Navigation Satellite System) from the 1950s to the 1980s induced low accuracy of ship position in single-beam surveys, leading to an uneven distribution of global seafloor topography quality

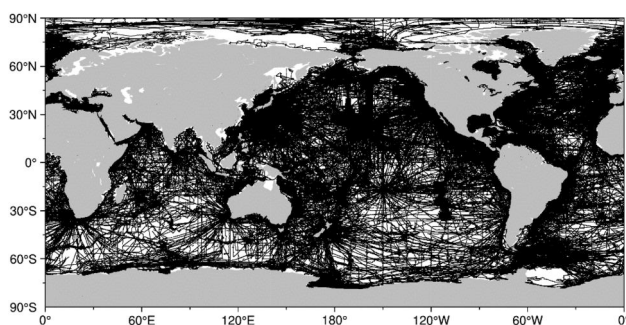


Figure 1. Global distribution of 5,464 single-beam track-lines from NCEI, including 119 million depth points.

(Smith 1993). Approximately 70% of the track-lines contain at least one erroneous data point (Becker et al. 2009). Therefore, it is necessary to evaluate and correct the global single-beam depths.

This study collected 5464 single-beam track-lines with a total of 119 million depth points from the National Centers for Environmental Information (NCEI). The accuracy of these ship-borne depths is evaluated by analyzing crossover errors and comparing them to the GEBCO_2023 model. Then, we classified the existing errors in the 5464 single-beam track-lines and developed a method to correct these errors. As a result, we obtained a new global single-beam dataset. The result can serve as a useful reference for the conservation and effective utilization of historical single-beam depths.

Data Sources

Most of the 5464 single-beam track-lines were collected from several oceanography research cruises conducted by a range of research organizations and governments around the world. Each track-line is assigned a unique source identification number (SID). [Figure 1](#) shows that there are approximately 119 million echo-sounding recordings in 5464 track-lines, but leaving gaps of hundreds kilometers between these lines (Mayer et al. 2018; Smith 1993).

[Figure 2](#) shows the numbers of single-beam echo sounder samples in each $1' \times 1'$ grid. Statistics show that about 9% of the 1 arc minute seafloor grids contain single-beam depth points. Therefore, the global distribution of single-beam depths is sparse and extremely heterogeneous. It indicates that there are more single-beam depths in the northern hemisphere than in the southern hemisphere. The region with relatively dense data is distributed near Japan in the western Pacific and along the west and east coasts of North America. The availability of single-beam data in the Indian Ocean and southern Pacific Ocean is quite limited. The scarcity of bathymetry data in the open ocean can be attributed to the slow speed of ships and fixed routes (Sandwell et al. 2006; Smith 1993). Furthermore, due to the sensitive data of the exclusive economic zones (Wöfl et al. 2019), the availability of bathymetry data in many coastal regions is similarly unsatisfactory.

[Figure 3](#) shows the accumulation of global single-beam depths collected each year up until 2017. The single-beam depths can be categorized into three distinct stages based on the mode of navigation (Smith 1993), including celestial navigation (before 1967), transit satellite (1967–1985) and GNSS (after 1985). The initial stage occurred prior to 1967, during which ships were generally positioned using celestial navigation techniques. The

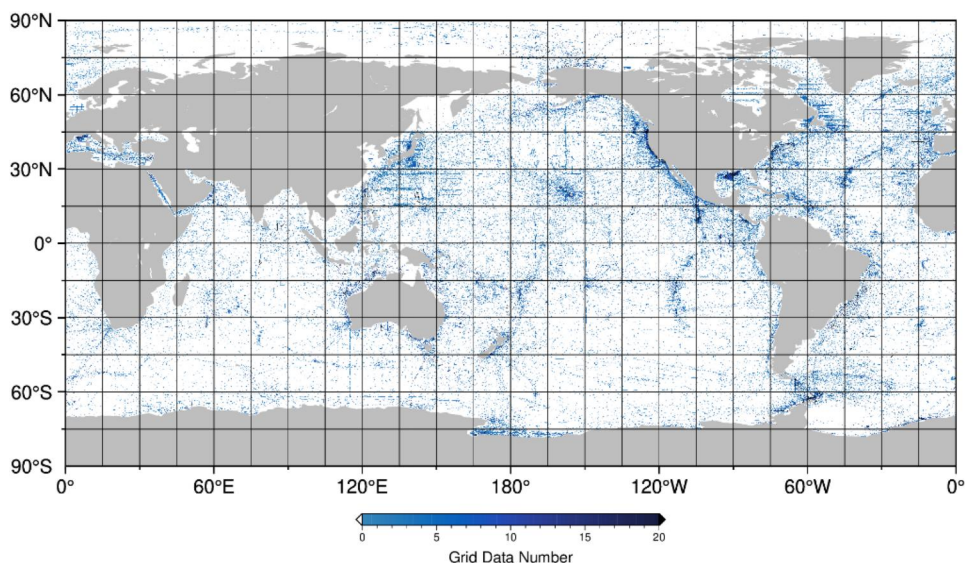


Figure 2. Numbers of echo-sounding samples in each $1' \times 1'$ grid. Statistics shows that about 9% of the seafloor has been sampled with single-beam echosounder. The number of single-beam echosounder samples in the Northern hemisphere is greater than that in the Southern hemisphere.

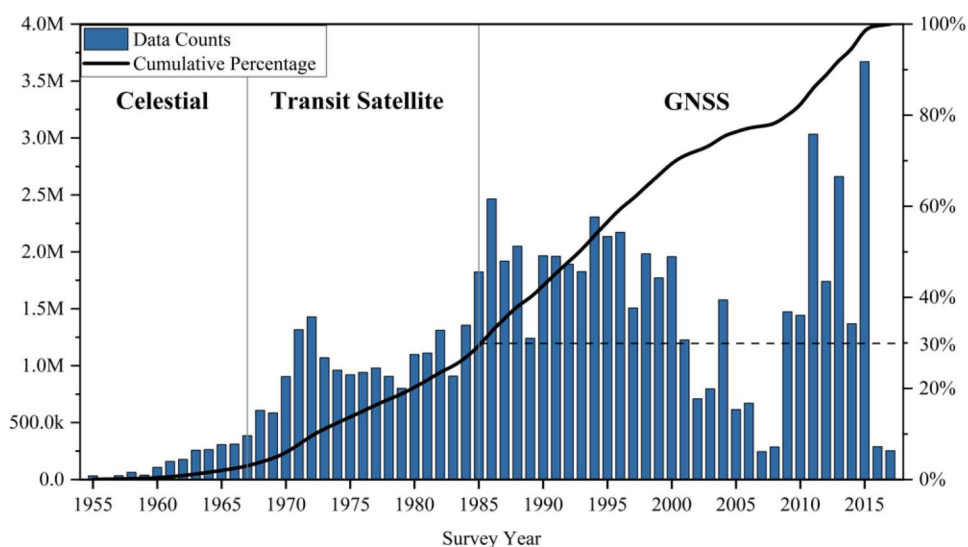


Figure 3. Cumulation of global single-beam depths collected each year up until 2017. The single-beam bathymetry data is categorized into three distinct stages based on the mode of navigation. Approximately 30% of the bathymetry data collected before 1985 did not use GNSS navigation. The majority of single-beam bathymetry data was obtained in 1970–2000.

second stage spans from 1967 to 1985, during which transit satellite navigation was commonly employed to determine the ship's location. The third stage, which began after 1985, witnessed the widespread adoption of the GNSS for the precise positioning of ships. Approximately 30% of the

bathymetry data collected before 1985 did not use GNSS navigation. The majority of single-beam bathymetry data was obtained in 1970–2000, with an annual average of about 1 million echo soundings. While the quantity of single-beam track-lines after 2000 is limited, the number of soundings exceeded 500,000 per year from 2010 to 2015, with the latest data collected in 2017. Furthermore, there are 21 million depth points in 787 single-beam track-lines without surveying time information, which are likely pre-1980s. The data recording format ‘MGD77’ containing time information began to be widely used in the 1980s (The marine geophysical data exchange format - MGD77T/Legacy MGD77, <https://www.ncei.noaa.gov/sites/default/files/2022-01/mgd77.pdf>). We speculate that the collection time of single-beam depth data with missing year information may have been earlier than 1980s.

Figure 4 shows the surveying time for the global single-beam data. Most of the single-beam data was collected in the twentieth century. However, in the twenty first century, scientists have obtained new data in the polar regions through scientific studies conducted in those areas (Dorschel et al. 2022; Jakobsson et al. 2020). The majority of the early data comes from the northern regions of the Pacific, Atlantic, and Indian Oceans. While some bathymetry data from unknown years is scattered in the Atlantic Ocean. Additionally, more recent single-beam data is distributed in the polar regions.

The GEBCO 2023 model was published in April 2023 and is a global terrain model for ocean and land, providing elevation data, in meters, on a 15

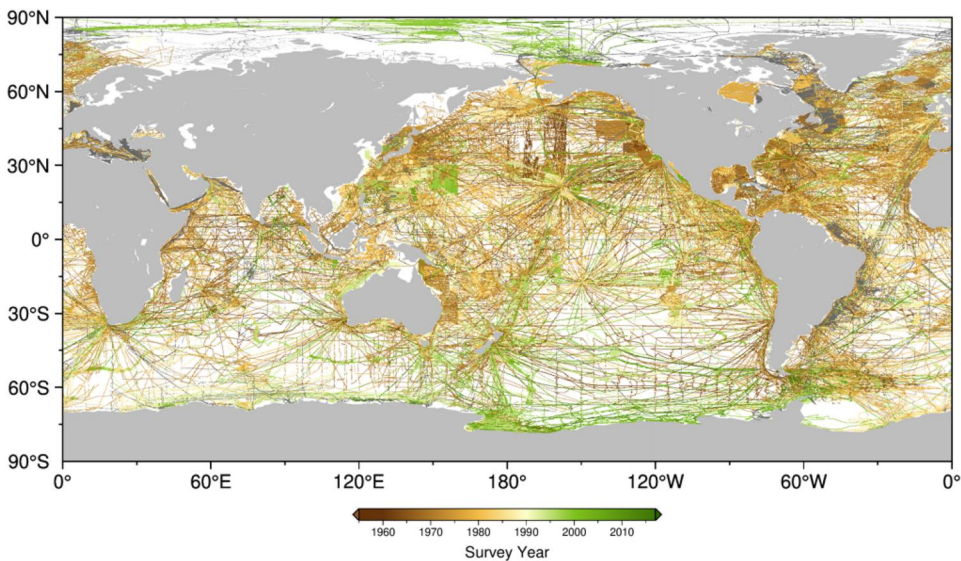


Figure 4. Global distribution of single-beam track-lines in surveying years. Most of the single-beam bathymetry data was collected in the twentieth century.

arc-second interval grid ((<https://www.gebco.net>). The grid uses as a 'base' Version 2.5.5 of the SRTM15+ data set (Tozer et al. 2019) between latitudes of 50° South and 60° North. This data set is a fusion of land topography with measured and estimated seafloor topography. It uses predicted depths based on the V32.1 gravity model (Sandwell et al. 2021). The SRTM15+ base grid is extended through a gridded depth measurement dataset developed by four Seabed 2030 Regional Centers (Mayer et al. 2018), mainly based on multibeam data, to produce the GEBCO_2023 model. All the data sources used in this paper were collected in Table 1.

Accuracy Evaluation

Given that global raw depth data contains a wide range of errors, Smith (1993) found through crossover error analysis that the most inaccurate depths are in the southern hemisphere ocean. Becker et al. (2009) compared the depths predicted from altimetric data with shipborne depths, and corrected the erroneous in ship data. Masetti et al. (2018) introduced triangulated irregular network (TIN) into nautical chart to achieve automatic correction of depth measurement data. This paper employs two methods to evaluate the accuracy of global single-beam depths. The first method is to analyze the crossover errors among the 5464 single-beam track-lines (Wessel 1989). This analysis provides information on the current accuracy of global single-beam data, including its distribution across the worldwide oceans, frequency, and statistical parameters. The second method involves calculating the difference between the global single-beam depth points and the GEBCO_2023 topography model.

External Crossover Errors

Crossover error (COE) analysis is one of the methods used to evaluate the accuracy of ship data (Wessel 1989). This method does not rely on pre-existing accurate measurements and instead determines acceptable values based on the majority principle (Wessel and Watts 1988). By searching for the intersection between two measurement track-lines, the linear interpolation method is used to obtain the depth value at the intersection point of the two track-lines, calculating the difference between them to get the COE. The COE is classified into internal crossover error (ICOE) and external crossover error (ECOE). ICOE is the COE of two track-lines that belong to the self-intersection of the track. ECOE is the COE among survey lines of different track-lines, which compares observations of different tracks. Smith (1993) evaluated the accuracy of the ship's bathymetry collected at the Lamont-Doherty Earth Observatory between 1955 and 1992.

The findings revealed that the most inaccurate data were observed in the Southern Ocean, with median external crossover errors ranging from 100 to 250 m (Smith 1993).

In order to evaluate the accuracy of the global single-beam depths, we use the x2sys module of GMT (Generic Mapping Tools) to calculate the ECOE of 5464 single-beam track-lines from NCEI (Wessel 2010). We processed each cruise individually. Each individual track-line was cross-referenced with others in the overlapping sea to ensure that no crossover points were missed. Then, we filtered out any duplicate points. We found 6.96 million external crossover points among 5464 single-beam track-lines. ECOEs in depth soundings are typically large and have long-tailed distributions, so need to be evaluated using a robust statistical parameter such as the median absolute deviation (MAD). The statistical analysis of 6.96 million external crossover points reveals that the MAD is 44.4 m, the mean error value is 377.7 m, and the standard deviation is 829.8 m (Table 2).

Figure 5a displays the global distribution of the 6.96 million ECOEs. It indicates that the single-beam depth's accuracy is generally higher in the northern hemisphere than that in the southern hemisphere, particularly in the northeastern Pacific and northwestern Indian Ocean. Seafloor with rapidly changing topography, such as the Mariana and Aleutian Trench in the Pacific, usually has higher errors. It is because errors in navigation may result in large inaccuracies in bathymetry measurements in regions with rough topography (Smith 1993; Wölfl et al. 2019). The larger ECOEs exist in the southern oceans, mostly due to the bad navigation of surveying ship. Celestial navigation in this area is challenging because the weather often obscures the sky for days, which leads to the collection of low-quality early single-beam bathymetry data at this specific area (Smith 1993). The updated bathymetry data with satellite navigation exhibited a significant difference compared to the previous data, resulting in greater ECOEs in this area.

Figure 5b illustrates the logarithmic histogram of the 6.96 million ECOEs. Approximately 5 million ECOEs have a difference of less than

Table 1. Data sources used in this paper.

Data sources	Descriptions	Processes	Data provider
Single-beam depths	~119 million points	Errors analysis and correction	NCEI
GEBCO_2023 Topography model	15 arc-second seafloor topography from multisource data	Accuracy evaluation of single-beam depths as a reference model	GEBCO

Table 2. The statistical of global single-beam ECOEs.

MAD(m)	MEAN(m)	STD(m)	Numbers of crossover points
44.4	377.7	829.8	6.96 million

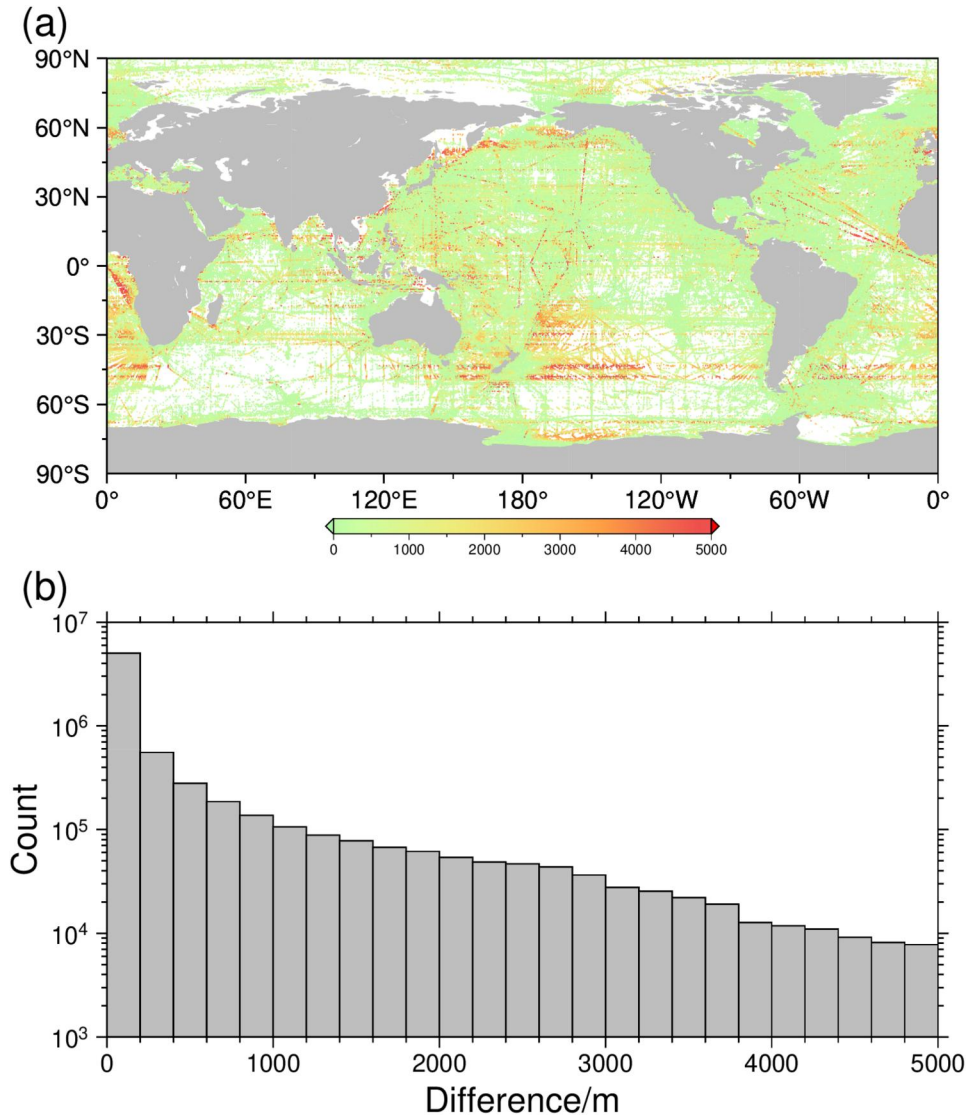


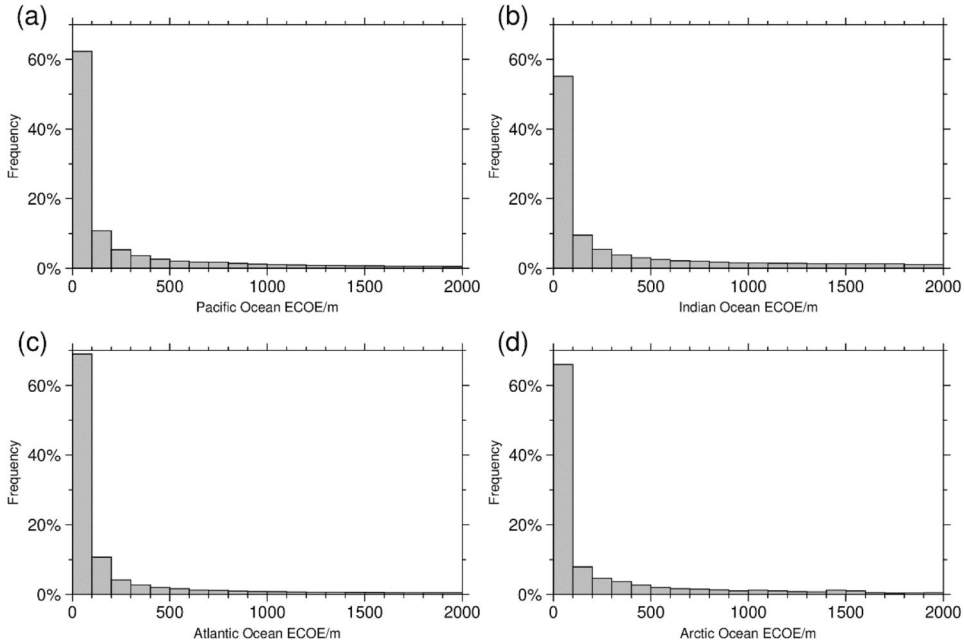
Figure 5. Global distribution (a) and logarithmic count (b) of 6.96 million ECOEs.

200 m, accounting for 72%. While 11.6% of the ECOEs exceed 1000 m, with a total of 0.81 million. The ECOEs reveal a significant presence of anomalous shipborne single-beam data around the world.

Table 3 provides statistics of the 6.96 million ECOEs. The global oceans are categorized into distinct regions: the Pacific Ocean (120°E – 100°W , -75°S – 70°N), the Indian Ocean (20°E – 120°E , -75°S – 26°N), the Atlantic Ocean (60°W – 20°E , -75°S – 0° ; 80°W – 0°E , 0° – 70°N), and the Arctic Ocean (70°N – 90°N). According to the median absolute deviation (MAD), the Atlantic Ocean has the highest quality bathymetry data, with a MAD of

Table 3. The statistics of single-beam ECOEs in world oceans.

Region	MAD(m)	MEAN(m)	STD(m)
Pacific Ocean	60.0	478.7	971.0
Indian Ocean	117.9	727.8	1154.3
Atlantic Ocean	34.1	312.7	723.5
Arctic Ocean	42.2	287.1	553.3

**Figure 6.** Histogram of the frequency distribution of ECOEs in the global oceans.

34.1 m. The Arctic and Pacific Oceans have MAD values of 42.2 and 60.0 m, respectively, the Indian Ocean has the lowest quality of bathymetry data, with a MAD of 117.9 m.

Figure 6 shows the histograms depicting the frequency distribution of ECOEs in the four oceans. About 67% of ECOEs in the Pacific Ocean have a value within 200 m, while the Indian Ocean has 56%, the Atlantic Ocean has 76%, and the Arctic Ocean has 72%. 15% of ECOEs in the Pacific Ocean have a value greater than 1000 m, compared to 25% in the Indian Ocean, 10% in the Atlantic Ocean, and 10% in the Arctic Ocean. These findings indicate that the Atlantic Ocean exhibits the highest data quality, while the Indian Ocean has the lowest. These results are consistent with the MAD statistics.

Difference from GEBCO_2023 Model

We compared the global single-beam depths with the GEBCO_2023 model. The differences between the single-beam points and the corresponding

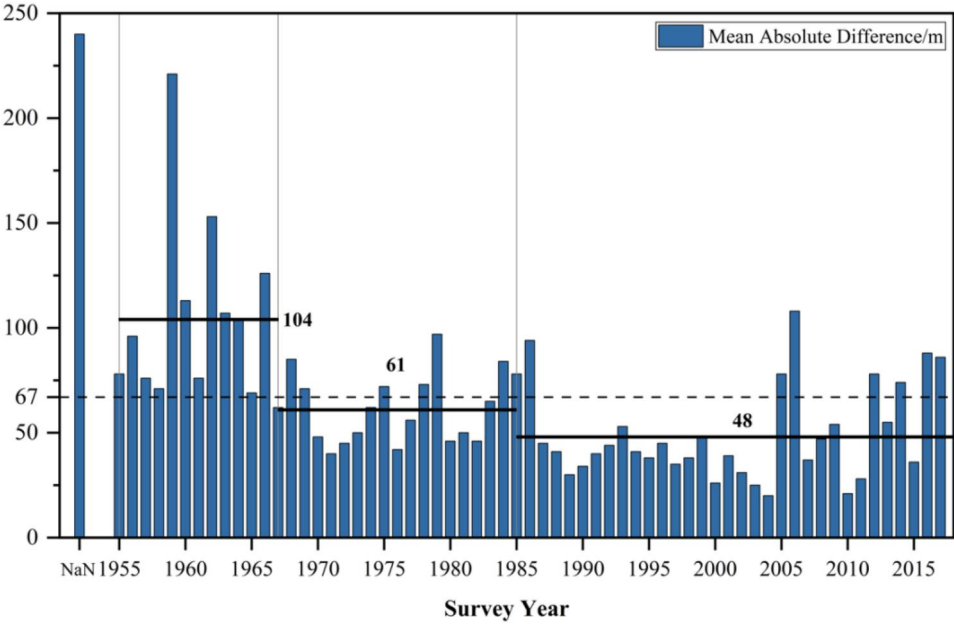


Figure 7. The average absolute discrepancy between the individual beam bathymetry measurements and the GEBCO_2023 model results for various years. The gradual improvement in the accuracy of single-beam bathymetry is mainly related to the advancement of ship navigation and positioning technology.

GEBCO_2023 grids' depth were calculated. Figure 7 shows the mean absolute differences between the single-beam depths and the GEBCO_2023 model for years, yielding an overall mean of 67 m. The gradual improvement of single-beam depths' accuracy is mostly due to the development of ship navigation and positioning technology. Before 1967, when celestial navigation was used, the mean absolute differences were typically approximately 100 m and exceeded 200 m in 1959. From 1967 to 1985, transit satellite navigation was commonly employed, resulting in enhanced accuracy in ship positioning. The mean absolute difference was improved to 61 m. The accuracy of depths improved significantly after 1985 due to the application of GNSS. After 1985, the mean absolute difference improved to 48 m. Moreover, there are 787 track-lines without time information, which show a mean absolute difference of 240 m. It indicates that there are two reasons for the large difference in these data. Firstly, the GEBCO_2023 model may not have used these data during the construction process. Secondly, the quality of these data may be very poor.

Table 4 presents the difference between the global single-beam depths and the GEBCO_2023 model. The minimum difference is -11683.0 m, the maximum difference is 10019 m, the mean difference is -54.1 m, the root

Table 4. The statistical of differences between the global single-beam and the GEBCO_2023 model.

MIN(m)	MAX(m)	MEAN(m)	RMS(m)	STD(m)	Numbers of Depth Points
-11683.0	10019.0	-54.1	261	260.9	119 million

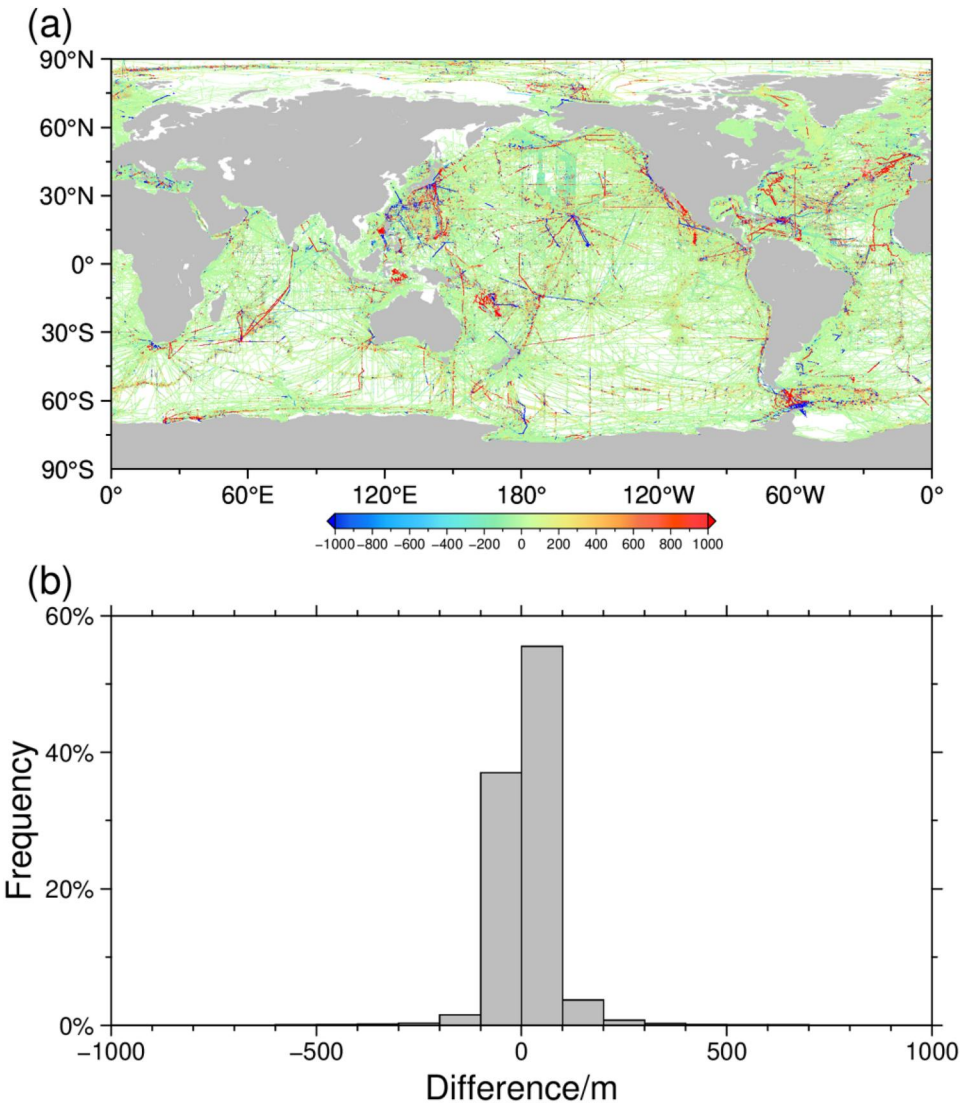


Figure 8. Distribution of the differences between the global single-beam depths and the GEBCO_2023 model (a). The frequency distribution histogram of the differences (b). Over 90% of the data points have a difference within 100 m.

mean square (RMS) variance is 261 m, and the standard deviation is 260.9 m.

Figure 8 shows the differences between the global single-beam depths and the GEBCO_2023 model. More than 90% of the data points have a

difference within 100 m. Most of the echo-sounding depths are shallower than the GEBCO_2023 model. This is because when the sonar propagates to the seabed within the coverage range, it returns the shallowest depth (Mayer et al. 2018). The GEBCO_2023 model also integrates multi-beam depths, resulting in a more refined seabed terrain. In the Arctic, GEBCO_2023 adopted the IBCAO dataset directly. The usage of multi-beam data (14.3%) is about 7 times that of single-beam data (1.9%) in IBCAO (Jakobsson et al. 2020). Thus, there are obviously differences between single-beam depths and the model in the Arctic.

There are distinct differences in some areas, such as the Arctic and the southwestern Indian Ridge (Robert et al. 1971), the Mariana Trench (Gardner et al. 2014), the Drake Passage (Bohoyo et al. 2019). In other areas with rough topography, small deviations in ship positioning may introduce large depth deviations, causing significant differences between single-beam depths and the model. In areas with relatively small terrain fluctuations, such as the eastern Pacific, a difference less than 100 m is acceptable. There is a strong correlation between the global distribution of differences and the distribution of typical ridge zones.

Errors Classify and Editing Methods

We used Python and GMT 6.5.0 (Wessel et al. 2019) to identify and correct errors in the single-beam depth dataset. The 5464 single-beam track-lines were stored in mgd77 format (Wessel and Chandler 2007), and each track-line has its own matching file. All the track-line files were converted to text files with three columns of longitude, latitude, and depth. Therefore, these files could be manipulated using standard UNIX commands such as awk, grep, and sed. GEBCO_2023 was used as a reference model to identify obvious errors in the single-beam depth dataset. Then, we classified types of errors and made necessary corrections to improve the single-beam depths dataset.

Classification of Errors

We developed three visualization windows specifically for facilitating data analysis (Figure 9). These windows include:

1. The statistic window shows differences between the ship depths and the GEBCO_2023 model.
2. The data window visualizes ship-borne and model depths along a track-line.

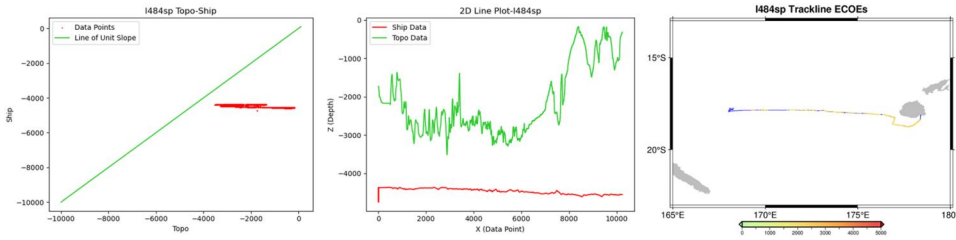


Figure 9. Three visualization windows specifically for facilitating data analysis. The l484sp cruise depth does not match the GEBCO_2023 model at all and the ECOEs of the l484sp cruise depth data on the entire survey are greater than 2000 m.

3. The map window depicts the geographic location and ECOEs of a ship track-line.

By examining the differences displayed in the statistics window, data window and map window, abnormal track-lines or observations can be identified and investigated.

Through analysis, we found 14 single-beam cruises are unavailable. For example, the l484sp cruise located in the South Pacific deviates from the GEBCO_2023 model significantly. As shown in Figure 9, there is poor consistency between the l484sp cruise depth data and GEBCO_2023 model in the statistical window. In the data window, the GEBCO_2023 model depths along the l484sp cruise show an obvious undulation of seafloor topography, while the single-beam soundings show a constant of about -4000 m. In the map window, the ECOEs of the l484sp cruise depth data on the entire survey are greater than 2000 m. Therefore, we contend that the l484sp cruise does not represent the true seafloor topography and is entirely impractical for use. Meanwhile, we have identified three types of errors:

1. Within the generally reliable ship track-lines, there are a few instances of significant errors in echo-sounding measurements, commonly referred to as outliers.
2. Errors arise from inaccuracies in converting sonar propagation time into depth, known as scale factor error.
3. The discrepancy in depth measurements caused by errors in recording the two-way propagation time of sonar is termed ‘travel time error’.

The outliers can be promptly eliminated, while the other two types of errors can be corrected.

Outliers

Most of the single-beam track-lines with significant errors are distributed in areas with rough topography (Figure 8), and the data quality in the southern hemisphere is worse than that in the northern hemisphere (Figure 5a). Therefore, we focus on single-beam depth analysis in these areas. The initial analysis concerns the correlation between the single-beam depth data and the GEBCO_2023 model, as determined through the application of a statistical window. Subsequently, the depth point is further corroborated with a difference of over 1000 m, as indicated by the data window, and duly marked. Finally, the depth point will be automatically removed if the ECOE value exceeds the specified threshold (1000 m) within the map window.

Figure 10 displays single-beam track-lines from the South Indian Ocean and South Atlantic Ocean. Figure 10a shows the tn058 cruise in the South Indian Ocean. The data window reveals the distribution of scattered anomalies (shown in red), with anomaly values varying from the model by more than 1000 m. Figure 10b shows the v1604 cruise in the South Atlantic Ocean. The statistics window indicates that there are several discrete data points (shown in red) distributed on both sides of a straight line of unit slope, and all of them differ from the model by more than 1000 m.

We systematically eliminated these outliers in the three cruises of tn058 and v1604 to improve the single-beam data quality. The accuracy statistics, both before and after eliminating the outliers, is presented in Table 5. The percentages of data elimination are 0.0001% and 0.0225%, respectively. The STD between single-beam depths and the GEBCO_2023 model has been

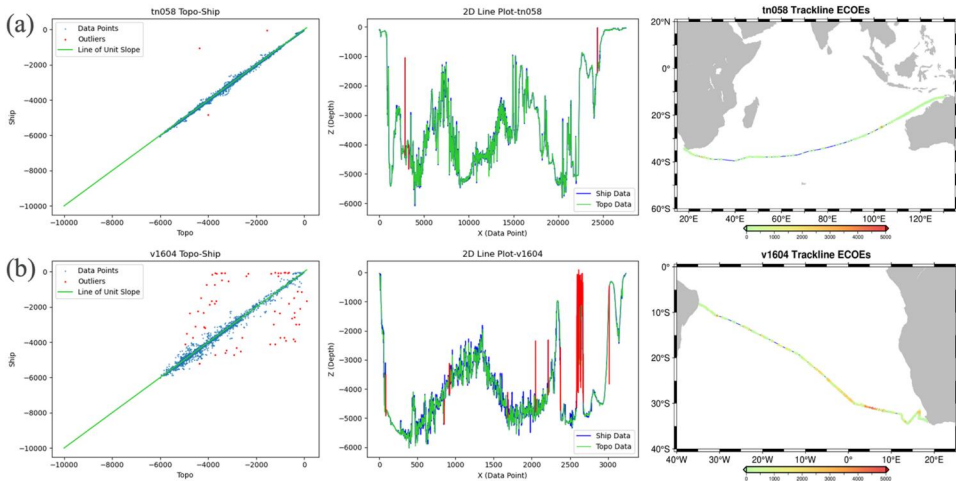


Figure 10. Visualization of single-beam and model depths along two track-lines: tn058 cruise in the South Indian Ocean area (a), v1604 cruise in the South Atlantic area (b). The significant outliers are shown with red points.

Table 5. Statistics of the differences between cruises of tn058, v1604 and GEBCO_2023 model, before and after errors correcting.

SID	tn058		v1604	
	Before	After	Before	After
MIN(m)	−816	435	−3537	−766
MAX(m)	3314	461	3676	683
MEAN(m)	−2	−1	−6	−2
STD(m)	34	26	331	128
Numbers of points	27561	27558	3234	3161

improved from 331 m to 128 m for v1604 cruise. Using the above method, we successfully eliminated outliers from a total of 2559 track-lines, resulting in an overall data removal rate of 0.02%.

Scale Factor Error

The scale factor error introduces overall deviations between the single-beam track-lines and the GEBCO_2023 model. As shown in [Figure 11](#), each depth is plotted as a point in the (x, y) plane, with the recorded value from the cruise on the y axis and the value interpolated by the GEBCO_2023 model on the x axis. In the analytical process, the linear fitting method is initially employed to calculate the appropriate slope between the recorded values from the cruise and the GEBCO_2023 model values. Then, underlying causes of the dimensionless ratio slope are investigated. Finally, the depth data is corrected by adjusting the slope. The dimensionless ratio slope should be close to 1 for the majority of the cruise depth data. A set of slope values was calculated for each cruise.

Thirty-four cruises exhibited a slope value which was different from unity. Four examples are shown in [Figure 11](#). [Figure 11a](#) shows a ship-topo fitting line slope of 0.75 for the 76010303 cruise. [Figure 11b](#) and [Figure 11d](#) shows ship-topo fitting line slope of 1.875 for mmw02 cruise and 0.533 for opr425rc cruise. [Figure 11c](#) shows a ship-topo fitting line slope of 0.547 for di91 cruise.

The scale factor error arose due to the wrong sound velocity unit being used when converting the echo-sounder time record to depth, as shown in [Table 6](#). The majority of the data pertaining to the 34 cruises with anomalous slopes is unavailable. However, the algorithm for correcting slopes can be employed to rectify the above cruises' depth data.

[Figure 12](#) displays the difference between modified depths from the four cruises and the GEBCO_2023 model. The modified depth data exhibits high consistency with the GEBCO_2023 model. [Table 7](#) presents the statistics of the difference between depths from the four cruises and the

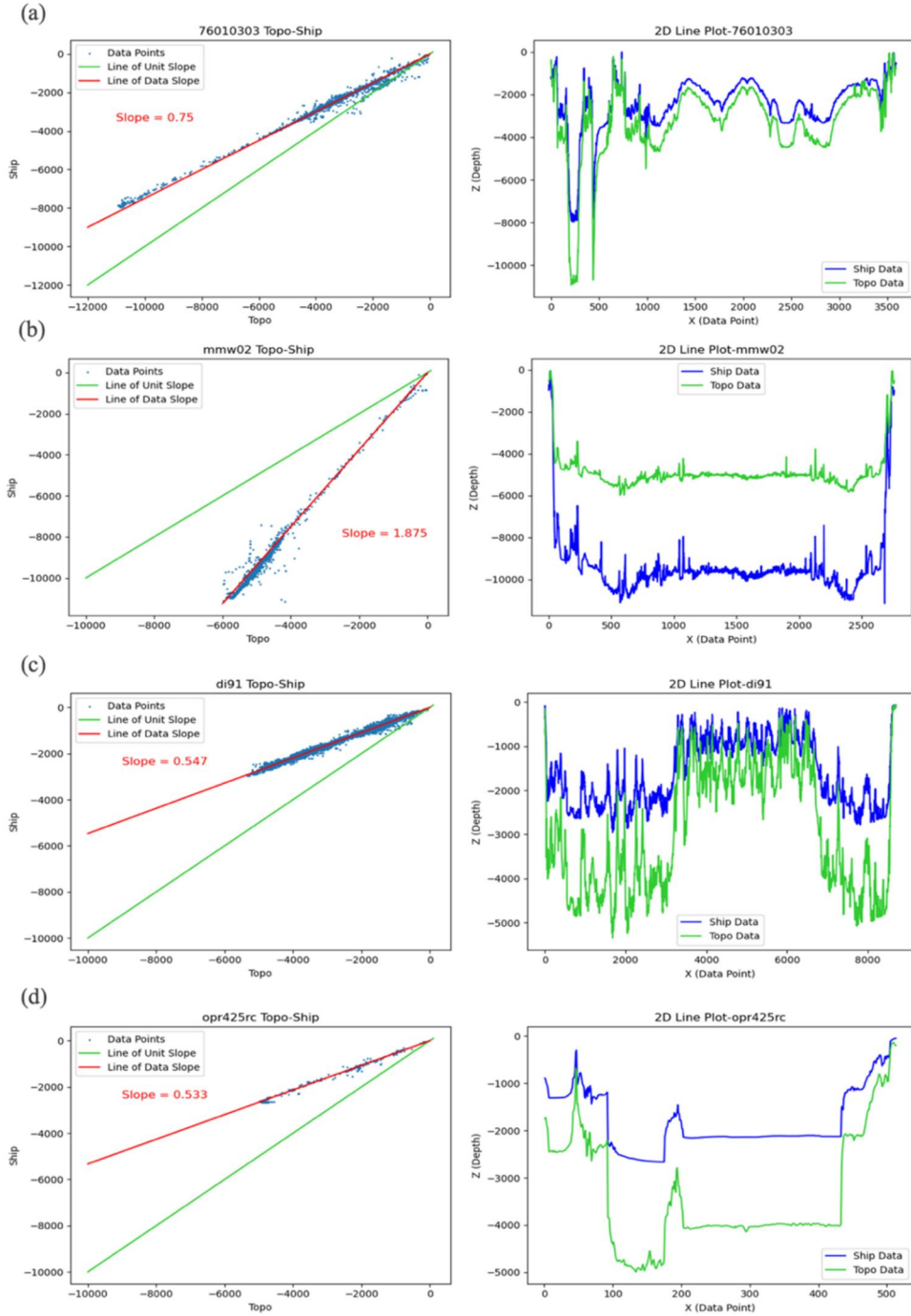


Figure 11. Scale factor errors are described by the four cruises, with the recorded value from the cruise on the y axis and the value interpolated by the GEBCO_2023 model on the x axis. (a) Ship-topo fitting line slope of 0.75 for 76010303 cruise. (b) Ship-topo fitting line slope of 1.875 for mmw02 cruise. (c) Ship-topo fitting line slope of 0.547 for di91 cruise. (d) Ship-topo fitting line slope of 0.533 for opr425rc cruise.

Table 6. Each special slope value and its possible reasons.

Slope	Possible reason
0.75	The relationship between sound velocity (1500 m/s) and two-way travel time (2000 ms)
1.875	The ratio of the sound velocity in meters (1500 m/s) to the speed in fathoms (800 fm/s)
0.547	The conversion from 1 fathom to 1 m
0.533	The ratio of the speed in fathoms (800 fm/s) to the sound velocity in meters (1500 m/s)

GEBCO_2023 model before and after scale factor errors were corrected. The mean differences between the two datasets improved from kilometers to tens of meters, and the standard deviation also improved significantly. Note that the slight outliers that remained after correction algorithm have been removed from 76010303 cruise and mmw02 cruise.

Travel Time Error

Travel time error arise because of an ambiguity in analog PDR records (Smith 1993). The PDR recorder controls and delays signals received from the sonar transducer using a circuit. It then moves a stylus on the chart paper using a rotating conveyor belt. However, due to rough terrain or poor sea conditions, some records on a trip may have noise or be missing, resulting in an error of N seconds in round trip time. The resulting water depth deviation (E) is:

$$E = V * N/2 \quad (1)$$

where E is the depth error, V is the sound velocity, typically taken as 1500 m/s, and N represents the two-way travel time error. Firstly, an automatic algorithm is employed to analyze whether a sequence of 750 integer multiples of the difference between the recorded values and the GEBCO_2023 model values exists. Subsequently, the travel error is confirmed through the sequence plot. Finally, the depth data is corrected by eliminate the travel error.

As shown in Figure 13, part of the depths in proto2mv cruise and pptu06wt cruise are biased by a well-fitting line (green line). Figure 14 shows the differences between ship depths and model along the two cruises. At points in the sequence of 0–300 and 1200–2200 along prot02mv cruise, the differences between ship depths and model are about 750 m. At points in the sequence of 100–600 along pptu06wt cruise, the differences between ship depths and model are about –1500 m. Such errors that are integer multiples of 750 m may be due to the ambiguity of the analogue PDR record of the single-beam system. When there is an error of N seconds recorded in the two-way travel time, it leads to $750 * N$ -meters bias in the depth measurements.

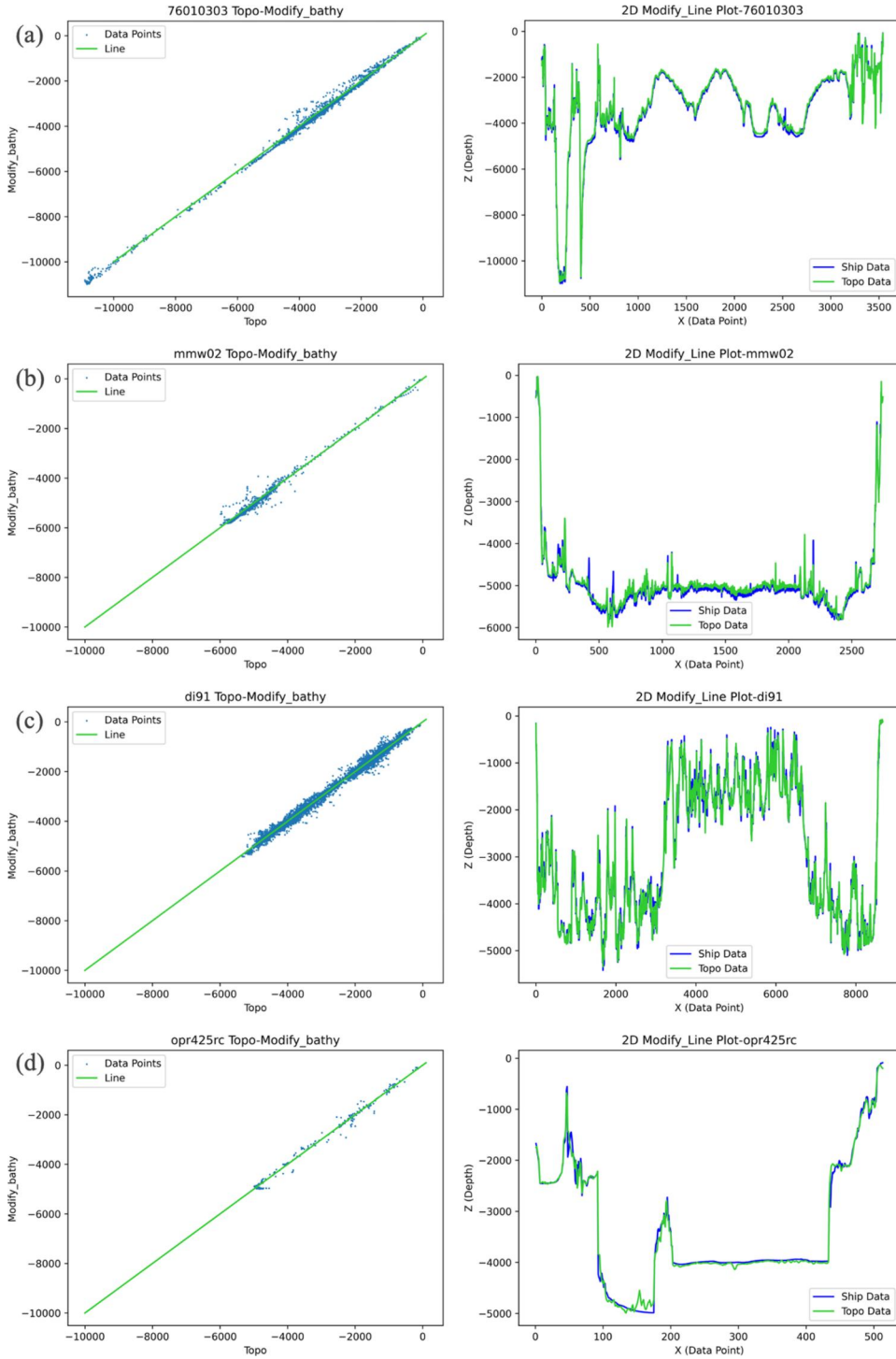


Figure 12. Correction results are described by the four cruise, with the modified value from the cruise on the y axis and the value interpolated by the GEBCO_2023 model on the x axis. The modified depth data exhibits high consistency with the GEBCO_2023 model.

Table 7. Statistics of the differences between cruises of 76010303, mmw02, di91 and opr425rc and the GEBCO_2023, before and after errors correcting.

SID	76010303		mmw02		di91		opr425rc	
	Before	After	Before	After	Before	After	Before	After
MEAN(m)	−826	63	4471	54	−1345	−12	−1581	−12
SLOPE	0.75	1.00	1.875	1.00	0.547	1.00	0.533	1.00
STD(m)	504	122	771	95	624	70	554	102
Numbers of points	3564	3543	2760	2745	8693	8693	513	513

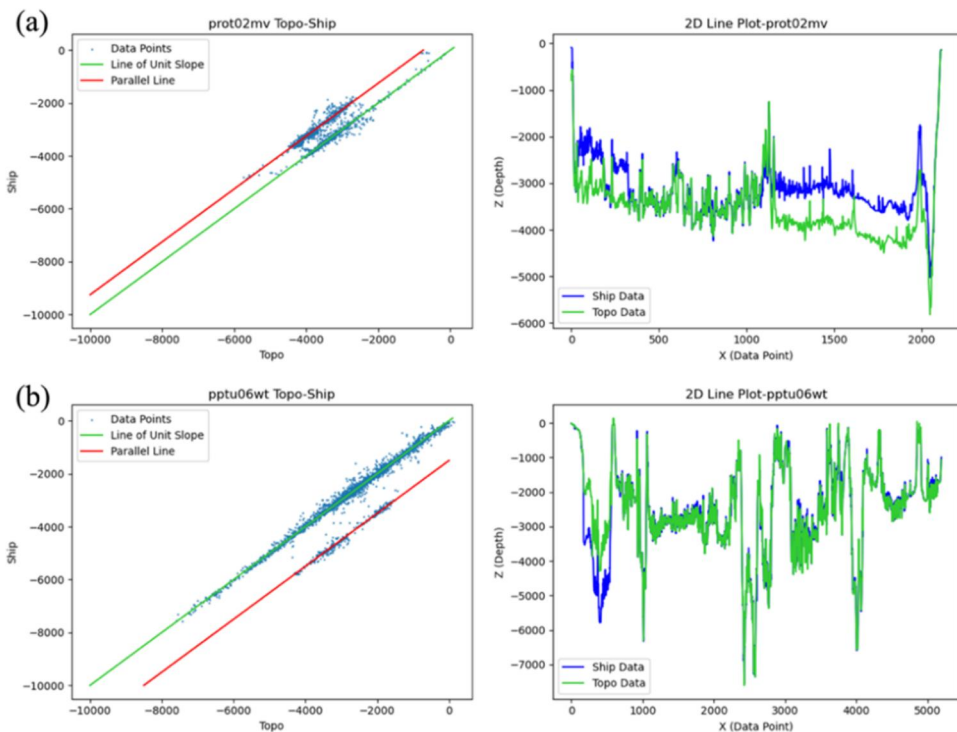


Figure 13. Part of depths in proto2mv cruise (a) and pptu06wt cruise (b) biased from a well-fitting line (green line). These errors may be due to the ambiguity of the analogue PDR record of the single-beam system.

We applied a correction of −750 m and 1500 m to the wrong points in proto2mv cruise and pptu06wt cruise, respectively. After rectifying these errors, the data quality improved significantly, and the mean and standard deviations of the differences between the two cruises and model decreased, as shown in Table 8. At last, we found and corrected 16 cruises with travel time errors.

Corrected Results

After analyzing the 5464 single-beam track-lines, it was found that:

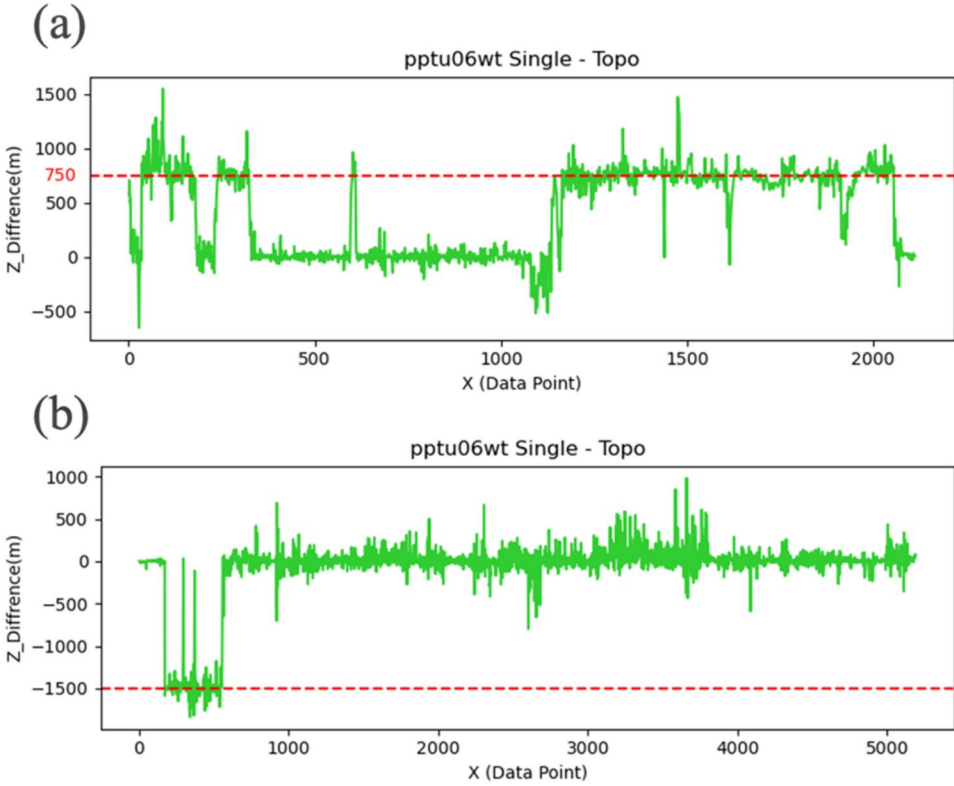


Figure 14. The differences between ship depths and model along the two cruises. At points in the sequence of 0–300 and 1200–2200 along prot02mv cruise, the differences between ship depths and model are about 750 m (a). At points in the sequence of 100–600 along ppter06wt cruise, the differences between ship depths and model are about -1500 m (b).

Table 8. Statistics of the differences between cruises of prot02mv and ppter06wt and the GEBCO_2023, before and after errors correcting.

SID	prot02mv		ppter06wt	
	Before	After	Before	After
MEAN(m)	−412	−21	85	−23
STD(m)	389	128	406	103
Numbers of points	2112	2046	5191	5150

1. 14 track-lines were identified with completely wrong depth records, including approximately 1.2 million points, which were deleted directly in this article.
2. 2559 track-lines exhibited obvious depth points' outliers. These outliers were detected and removed in this article, about 1.0 million points were deleted in total.
3. 34 track-lines presented scale factor errors. These track-lines were corrected by using an algorithm to derive a right scale factor and replace the old one.

Table 9. Statistics of differences between ship depths and GEBCO_2023, before and after errors correction.

Type	Outliers		Scale factor errors		Travel time errors	
Numbers of trackline	2559		34		16	
	Before	After	Before	After	Before	After
MEAN(m)	-10	-6	-1606	123	-1696	4
STD(m)	257	92	949	80	1887	160
Numbers of points (million)	47.8	46.8	0.528	0.513	1.46	1.44

- 16 track-lines displayed travel time errors. All the error points were detected and corrected by adding an integer multiple of 750 m.

In addition, there were 2841 track-lines of good quality that were not processed in this article.

Table 9 presents the statistics of differences between ship depths and model, before and after error correction. By removing approximately 1 million abnormal data points from the 2559 track-lines, the mean difference decreased from -10 m to -6 m, and the standard deviation decreased from 257 m to 92 m. By correcting the scale factor of 34 track-lines, the mean difference decreased from -1606 m to 123 m, and the standard deviation was reduced from 949 m to 80 m. By correcting the travel time errors of about 0.888 million abnormal points in 16 track-lines, the mean difference decreased from -1696 m to 4 m, and the standard deviation was reduced from 1887 m to 160 m.

After data correction, this article obtained a total of 115 million echosounding points from 5450 single-beam track-lines, with an overall data elimination rate of about 4%. About 1.4 million points were corrected and retained. The corrected global single-beam dataset was compared with the GEBCO2023 model, and the difference is shown in Figure 15. Most of the differences between the corrected single-beam depths and the model are within 200 m, accounting for approximately 99%. The statistics of differences between the new single-beam dataset and the GEBCO_2023 model are shown in Table 10. It indicates that the mean differences decreased from -54.1 m to 6.8 m, and the standard deviation decreased from 260.9 m to 68.5 m. Obviously, the data quality has improved significantly.

Discussion

Despite the algorithm's evaluation and correction of global single-beam data, a minor proportion of uncorrected errors may remain due to the robust correction method employed. In the case of the first type of outlier, it is possible that outlier points may have been identified incorrectly. This discrepancy can be attributed to the fact that the GEBCO2023 model did

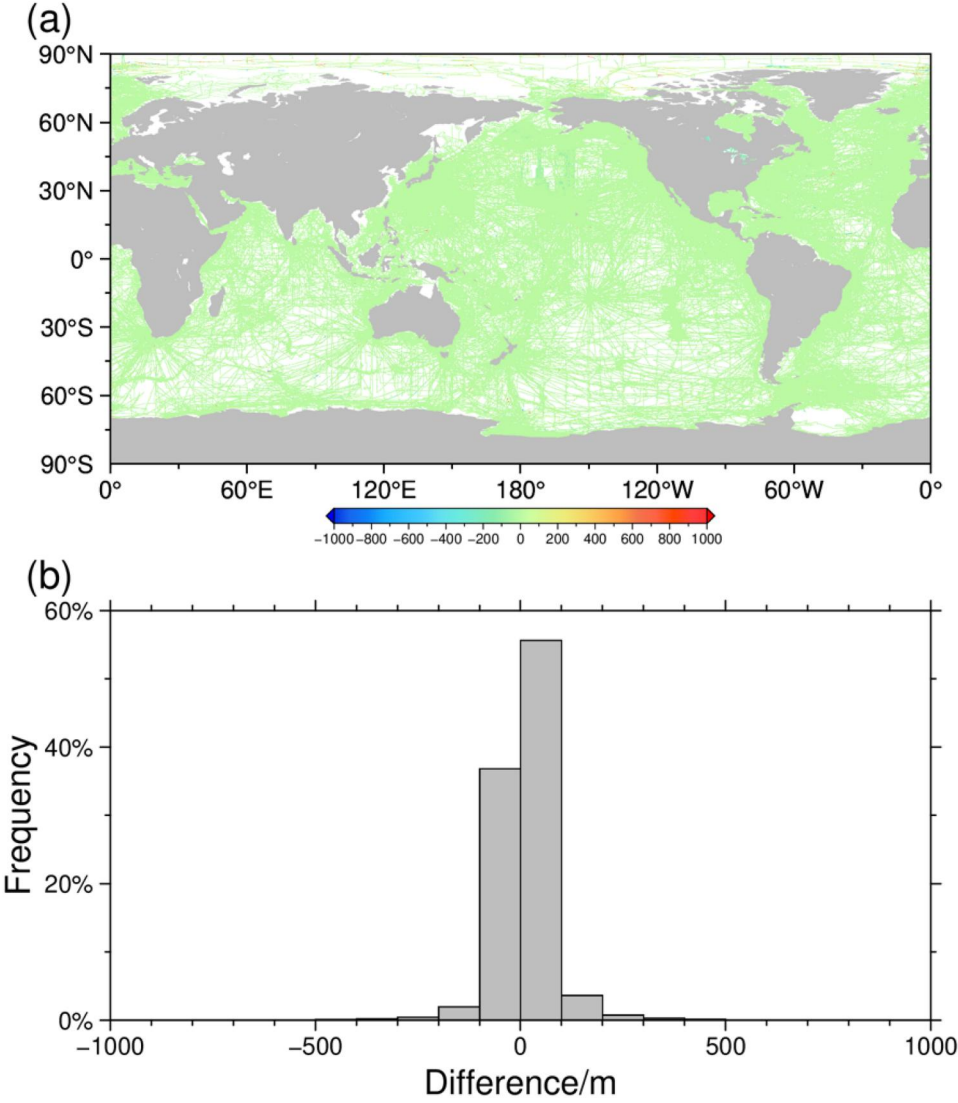


Figure 15. Global distribution of the difference between the new single-beam dataset and GEBCO_2023 (a) and the frequency distribution histogram. The differences between the corrected single-beam depths and the GEBCO_2023 model are within 200 m, accounting for approximately 99%.

Table 10. The statistics of differences between the new single-beam dataset and GEBCO_2023 (refer to Table 4).

MIN (m)	MAX (m)	MEAN (m)	RMS (m)	STD (m)	Numbers of depth points
-998.0	999.0	6.8	68.9	68.5	115 million

not combine these data for inversion, resulting in a significant discrepancy between the depth value and the model value. Regarding the second and third types, the majority of data points in the 50 track-lines are unavailable (with significant errors), based on the reliable accuracy of GEBCO2023 and

their special patterns with the model. Consequently, we have elected to implement corrections, despite the potential risks involved.

The velocity of sound in water is subject to variation due to salinity, temperature, and pressure, resulting in a difference of 2.5% between the two commonly used nominal velocities (1463 m/s and 1500 m/s) (Carter 1980). This minor deviation is challenging to detect in data comparison. This research method makes use of depth information exclusively and does not take in account factors such as the location of the measurement and changes in sound velocity. More approaches need to be combined to enhance the precision of single-beam data. Sound Speed Manager provides a streamlined workflow that guides the users to perform accurate processing and management of sound speed profiles (Masetti et al. 2017). A ray-tracing uncertainty estimation tool is performed by spatial variability analysis at the specific time of validity of the ocean model products, with the potential applications in ocean mapping (Masetti et al. 2018).

Conclusion

Global single-beam depths are crucial for comprehending the seafloor topography. During the ship navigation positioning procedure, data recording, and historical data format conversion, there are inevitably some errors and outliers, leading to large discrepancies in certain data. This paper conducts a comprehensive analysis of the accuracy of the global 5464 single-beam track-lines from NCEI. The errors in this dataset were detected and corrected to improve the quality of data. The work presented in this paper is highly beneficial in rescuing valuable historical observation data and facilitating its practical application.

The analysis of ECOEs reveals that the Atlantic Ocean has the highest quality of sounding data, while the Indian Ocean has the lowest quality. About 11.6% of the ECOEs exhibit errors exceeding 1000 m. By comparing the global single-beam depths with the GEBCO_2023 model, the accuracy of single-beam depths gradually improved as ship navigation and positioning technology advanced, reducing the error from 104 m in celestial navigation to 48 m in GNSS navigation. Currently, approximately 48% of the publicly available single-beam track-lines contain at least one inaccurate data point. We developed a visualization tool using Python and GMT to help data analysis. We found and classified unavailable track-lines and three kinds of errors, including outliers, scale factor errors, and travel time errors. Following correction, the accuracy of global single-beam depth data has been significantly enhanced.

Although the algorithm has corrected global single-beam data, the necessity for higher precision multibeam data and more sophisticated methods

remains to analyze potential residual errors in single-beam data. The uncertainty of sonar detection is only a possible source of error, including post depth analysis and processing, data integration and reduction, which may seriously affect the final survey soundings (Hughes Clarke 2018). The International Hydrographic Organization (IHO) has developed rules and best practices for conducting hydrographic surveys to further improve the quality of survey depth data (International Hydrographic Bureau 2008). In the future, new technologies for ocean surveying and mapping will be applied, including autonomous data collection systems (Wynn et al. 2014), automated data processing systems (Rezvani et al. 2015), and others. The accumulation of data from the global ocean will continue to increase in depth data quality.

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Disclosure Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability Statement

Python and GMT code files can be used to analyze and correct single beam depth data, are available at <https://github.com/lx010100/Analysis-Tool.git>.

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